COMPLETE SOURCE CODE EXPLANATION

Services & Multi-Threading Engine

Line-by-Line Explanation of Business Logic & Background Threads

In-depth analysis of 15 Service classes. Highlights Multi-Threading with KitchenOrderProcessor, DatabaseHealthMonitor, AsyncReportGenerator, and business rule orchestration.

Project: Restaurant Management System

Language: Java 17 / JavaFX / JDBC

Scope: Detailed Line-by-Line Breakdown

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AsyncReportGenerator.java

Path: src\main\java\com\restaurant\service\AsyncReportGenerator.java | Total Lines: 199 | Logical Blocks: 7

📘 COMPONENT OVERVIEW & RESPONSIBILITY

Defines and executes the AsyncReportGenerator role in the Restaurant Management architecture.

Lines 1–37 Package & Library Imports

```
package com.restaurant.service;

import com.restaurant.model.Order;
import com.restaurant.util.ActivityLogger;
import com.restaurant.util.ReportExporter;

import java.io.IOException;
import java.util.List;
import java.util.Map;
import java.util.concurrent.CompletableFuture;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
import java.util.function.Consumer;
import java.util.logging.Level;
import java.util.logging.Logger;

/**
 * MULTI-THREADING — Async Report Generator
 *
 * Wraps time-consuming report queries + CSV exports inside CompletableFuture
 * so they run on a background thread pool and never block the JavaFX UI thread.
 *
 * Demonstrates:
 *  - CompletableFuture.supplyAsync() with a custom ExecutorService
 *  - thenAccept() for UI callback on the FX thread via Platform.runLater
 *  - exceptionally() for clean async error handling
 *  - ExecutorService (fixed thread pool) management
 *
 * Usage example (from a Controller):
 *
 *   AsyncReportGenerator.getInstance().exportOrdersAsync(
 *       orders,
 *       "daily_orders",
 *       path -> showAlert("Saved to: " + path),
 *       err  -> showAlert("Export failed: " + err.getMessage())
 *   );
 */
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Concurrency Support	Multi-threading utilities (BlockingQueue, Executors, Atomic primitives) used for asynchronous tasks.
File Handling I/O	File streams and writers (FileWriter, BufferedWriter, PrintWriter) for receipts and log generation.
Line Analysis	Covers lines 1 to 37. Cleanly separates dependencies without wildcard star imports where specificity is required.

`Lines 38–38` **Class Declaration & Heritage**

```
38  public class AsyncReportGenerator {
```

Architecture Role	Defines AsyncReportGenerator as a core component in the Restaurant Management System.
Domain Entity	Encapsulates business entity attributes, immutable identity, and state transformations.
Access & Modifiers	Declared with public visibility on line 38, allowing cross-package invocation.

Lines 39–97 **Class Members & Implementation**

```
39
40      private static final Logger LOGGER =
41              Logger.getLogger(AsyncReportGenerator.class.getName());
42
43      // Singleton
44      private static AsyncReportGenerator instance;
45
46      /** 2-thread pool: one for reports, one for CSV export running in parallel if needed. */
47      private final ExecutorService executor =
48              Executors.newFixedThreadPool(2, runnable -> {
49                  Thread t = new Thread(runnable, "Report-Generator-Thread");
50                  t.setDaemon(true);
51                  return t;
52              });
53
54      private final ReportExporter reportExporter = new ReportExporter();
55      private final ReportService  reportService  = new ReportService();
56
57      // ------------------------------------------------------------------------
58      // Singleton
59      // ------------------------------------------------------------------------
60
61      private AsyncReportGenerator() {}
62
63      public static synchronized AsyncReportGenerator getInstance() {
64          if (instance == null) {
65              instance = new AsyncReportGenerator();
66          }
67          return instance;
68      }
69
70      // ------------------------------------------------------------------------
71      // Async export methods
72      // ------------------------------------------------------------------------
73
74      /**
75       * Exports a list of orders to CSV asynchronously.
76       *
77       * @param orders      order data to export
78       * @param filename    base filename (no extension)
79       * @param onSuccess   callback with the saved file path — called when done
80       * @param onError     callback with the exception — called on failure
81       * @return CompletableFuture so callers can chain more actions if needed
82       */
83      public CompletableFuture<String> exportOrdersAsync(
84              List<Order> orders,
85              String filename,
86              Consumer<String> onSuccess,
87              Consumer<Throwable> onError) {
88
89          return CompletableFuture
90                  .supplyAsync(() -> {
91                      // Runs on Report-Generator-Thread (NOT the JavaFX thread)
92                      try {
93                          LOGGER.info("AsyncReportGenerator: starting order CSV export...");
94                          String path = reportExporter.exportOrdersToCSV(orders, filename);
95                          ActivityLogger.getInstance().logReportExported("ORDERS_CSV", path);
96                          return path;
97                      } catch (IOException e) {
```

Component	Class Members & Implementation
Multi-Threading Logic	Manages worker threads, background queues, or concurrent task execution without blocking the JavaFX UI rendering loop.
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Thread Synchronization	Guarantees thread-safety: prevents race conditions by allowing only one thread to execute this critical block at any instant.
Line-by-Line Breakdown	Lines 39–97 (59 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 98–142 Class Members & Continuation

```
                        throw new RuntimeException("Failed to export orders CSV: " + e.getMessage(),
e);
                    }
                }, executor)
                .thenApply(path -> {
                    if (onSuccess != null) {
                        // Delegate back to caller — caller should wrap in Platform.runLater if JavaFX
                        onSuccess.accept(path);
                    }
                    return path;
                })
                .exceptionally(ex -> {
                    LOGGER.log(Level.SEVERE, "AsyncReportGenerator: export failed", ex);
                    ActivityLogger.getInstance().logError("AsyncReportGenerator", ex.getMessage());
                    if (onError != null) {
                        onError.accept(ex);
                    }
                    return null;
                });
    }

    /**
     * Fetches the sales-by-category report AND exports it to CSV, all asynchronously.
     *
     * @param onSuccess callback with the saved file path
     * @param onError   callback with the exception on failure
     * @return CompletableFuture for optional chaining
     */
    public CompletableFuture<String> exportSalesByCategoryAsync(
            Consumer<String> onSuccess,
            Consumer<Throwable> onError) {

        return CompletableFuture
                .supplyAsync(() -> {
                    try {
                        LOGGER.info("AsyncReportGenerator: fetching sales by category...");
                        // DB query — can be slow, runs on background thread
                        Map<String, Double> data = reportService.getSalesByCategory();

                        LOGGER.info("AsyncReportGenerator: exporting sales CSV...");
                        String path = reportExporter.exportSalesByCategoryCSV(data);
                        ActivityLogger.getInstance().logReportExported("SALES_BY_CATEGORY", path);
                        return path;
                    } catch (IOException e) {
                        throw new RuntimeException("CSV export failed: " + e.getMessage(), e);
                    }
```

Component	Class Members & Continuation
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 98–142 (45 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 143–188 Class Members & Continuation

```
143                 }, executor)
144                 .thenApply(path -> {
145                     if (onSuccess != null) onSuccess.accept(path);
146                     return path;
147                 })
148                 .exceptionally(ex -> {
149                     LOGGER.log(Level.SEVERE, "AsyncReportGenerator: sales category export failed", e
x);
150                     ActivityLogger.getInstance().logError("AsyncReportGenerator", ex.getMessage());
151                     if (onError != null) onError.accept(ex);
152                     return null;
153                 });
154     }
155 
156     /**
157      * Exports the daily items breakdown to CSV asynchronously.
158      *
159      * @param orders    today's orders (pre-fetched from OrderService)
160      * @param onSuccess callback with the saved file path
161      * @param onError   callback with the exception on failure
162      * @return CompletableFuture for optional chaining
163      */
164     public CompletableFuture<String> exportDailyItemsAsync(
165             List<Order> orders,
166             Consumer<String> onSuccess,
167             Consumer<Throwable> onError) {
168 
169         return CompletableFuture
170                 .supplyAsync(() -> {
171                     try {
172                         LOGGER.info("AsyncReportGenerator: exporting daily items CSV...");
173                         String path = reportExporter.exportDailyItemsCSV(orders);
174                         ActivityLogger.getInstance().logReportExported("DAILY_ITEMS", path);
175                         return path;
176                     } catch (IOException e) {
177                         throw new RuntimeException("Daily items export failed: " + e.getMessage(), e);
178                     }
179                 }, executor)
180                 .thenApply(path -> {
181                     if (onSuccess != null) onSuccess.accept(path);
182                     return path;
183                 })
184                 .exceptionally(ex -> {
185                     LOGGER.log(Level.SEVERE, "AsyncReportGenerator: daily items export failed", ex);
186                     if (onError != null) onError.accept(ex);
187                     return null;
188                 });
```

Component	Class Members & Continuation
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 143–188 (46 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 189–194` Class Members & Continuation

```
189         }
190
191         /**
192          * Shuts down the executor service cleanly.
193          * Should be called in Application.stop() or on application exit.
194          */
```

Component	Class Members & Continuation
Line-by-Line Breakdown	Lines 189–194 (6 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 195–199` Method: shutdown()

```
195         public void shutdown() {
196             executor.shutdown();
197             LOGGER.info("AsyncReportGenerator executor shut down.");
198         }
199     }
```

Component	Method: shutdown()
Line-by-Line Breakdown	Lines 195–199 (5 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

AuthenticationService.java

Path: src\main\java\com\restaurant\service\AuthenticationService.java | Total Lines: 79 | Logical Blocks: 8

📘 COMPONENT OVERVIEW & RESPONSIBILITY

Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–16 Package & Library Imports

```
1   package com.restaurant.service;
2
3   import com.restaurant.data.DataStructureManager;
4   import com.restaurant.enums.UserRole;
5   import com.restaurant.exception.InvalidLoginException;
6   import com.restaurant.model.User;
7   import com.restaurant.repository.UserRepository;
8   import com.restaurant.util.ActivityLogger;
9   import com.restaurant.util.PasswordUtil;
10  import com.restaurant.util.ValidationUtil;
11
12  import java.util.logging.Logger;
13
14  /**
15   * Service managing user authentication, sessions, and role verification.
16   */
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Line Analysis	Covers lines 1 to 16. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 17–17 Class Declaration & Heritage

```
17  public class AuthenticationService {
```

Architecture Role	Defines AuthenticationService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 17, allowing cross-package invocation.

Lines 18–32 Class Members & Implementation

```
18      private static final Logger LOGGER = Logger.getLogger(AuthenticationService.class.getName());
19      private final UserRepository userRepository;
20      private static User currentUser;
21
22      public AuthenticationService() {
23          this.userRepository = new UserRepository();
24      }
25
26      public AuthenticationService(UserRepository userRepository) {
27          this.userRepository = userRepository;
28      }
29
30      /**
31       * Authenticates user against stored salted SHA-256 hash.
32       */
```

Component	Class Members & Implementation
Line-by-Line Breakdown	Lines 18–32 (15 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 33–57

Method: login()

```
33      public User login(String username, String plainPassword) throws InvalidLoginException {
34          if (!ValidationUtil.isNotEmpty(username) || !ValidationUtil.isNotEmpty(plainPassword)) {
35              throw new InvalidLoginException("Username and password must not be empty.");
36          }
37
38          User user = userRepository.findByUsername(username.trim());
39          if (user == null) {
40              LOGGER.warning("Authentication failed: User not found for username '" + username + "'");
41              throw new InvalidLoginException("Invalid username or password.");
42          }
43
44          boolean match = PasswordUtil.verifyPassword(plainPassword, user.getPasswordHash(), user.getSal
t());
45          if (!match) {
46              LOGGER.warning("Authentication failed: Incorrect password for username '" + username +
"'");
47              throw new InvalidLoginException("Invalid username or password.");
48          }
49
50          currentUser = user;
51          DataStructureManager.getInstance().registerSession(user.getUsername());
52          LOGGER.info("User '" + username + "' logged in successfully as " + user.getRole());
53          // FILE HANDLING — write login event to activity log
54          ActivityLogger.getInstance().logLogin(user.getUsername(), user.getRole().name());
55          return user;
56      }
57
```

Component	Method: login()
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 33–57 (25 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 58–67

Method: logout()

```
58      public void logout() {
59          if (currentUser != null) {
60              LOGGER.info("User '" + currentUser.getUsername() + "' logged out.");
61              // FILE HANDLING — write logout event to activity log
62              ActivityLogger.getInstance().logLogout(currentUser.getUsername());
63              DataStructureManager.getInstance().endSession(currentUser.getUsername());
64              currentUser = null;
65          }
66      }
67
```

Component	Method: logout()
Line-by-Line Breakdown	Lines 58–67 (10 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 68–71 Method: getCurrentUser()

```
68      public static User getCurrentUser() {
69          return currentUser;
70      }
71
```

Component	Method: getCurrentUser()
Line-by-Line Breakdown	Lines 68–71 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 72–75 Method: isLoggedIn()

```
72      public static boolean isLoggedIn() {
73          return currentUser != null;
74      }
75
```

Component	Method: isLoggedIn()
Line-by-Line Breakdown	Lines 72–75 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 76–79 Method: hasRole()

```
76      public static boolean hasRole(UserRole role) {
77          return currentUser != null && currentUser.getRole() == role;
78      }
79  }
```

Component	Method: hasRole()
Line-by-Line Breakdown	Lines 76–79 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

BillingService.java

Path: src\main\java\com\restaurant\service\BillingService.java | Total Lines: 119 | Logical Blocks: 9

> 📘 **COMPONENT OVERVIEW & RESPONSIBILITY**
>
> Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–16 Package & Library Imports

```
1  package com.restaurant.service;
2
3  import com.restaurant.config.AppConfig;
4  import com.restaurant.model.Order;
5  import com.restaurant.model.OrderItem;
6  import com.restaurant.util.ActivityLogger;
7  import com.restaurant.util.BillReceiptWriter;
8  import com.restaurant.util.CurrencyUtil;
9  import com.restaurant.util.DateTimeUtil;
10
11 import java.io.IOException;
12 import java.util.List;
13
14 /**
15  * Service dedicated to bill calculations, discounts, taxes, and receipt generation.
16  */
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
File Handling I/O	File streams and writers (FileWriter, BufferedWriter, PrintWriter) for receipts and log generation.
Line Analysis	Covers lines 1 to 16. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 17–17 Class Declaration & Heritage

```
17 public class BillingService {
```

Architecture Role	Defines BillingService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 17, allowing cross-package invocation.

Lines 18–32 Class Members & Implementation

```
18
19      private final double taxRate;
20      private final double serviceChargeRate;
21
22      public BillingService() {
23          AppConfig config = AppConfig.getInstance();
24          this.taxRate = config.getDouble("app.taxRate", 5.0) / 100.0;
25          this.serviceChargeRate = config.getDouble("app.serviceChargeRate", 2.5) / 100.0;
26      }
27
28      public BillingService(double taxPercentage, double serviceChargePercentage) {
29          this.taxRate = taxPercentage / 100.0;
30          this.serviceChargeRate = serviceChargePercentage / 100.0;
31      }
32
```

Component	Class Members & Implementation
Line-by-Line Breakdown	Lines 18–32 (15 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 33–37 Method: calculateSubtotal()

```
33      public double calculateSubtotal(List<OrderItem> items) {
34          if (items == null) return 0.0;
35          return items.stream().mapToDouble(OrderItem::getSubtotal).sum();
36      }
37
```

Component	Method: calculateSubtotal()
Line-by-Line Breakdown	Lines 33–37 (5 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 38–41 Method: calculateTax()

```
38      public double calculateTax(double taxableAmount) {
39          return Math.max(0, taxableAmount) * taxRate;
40      }
41
```

Component	Method: calculateTax()
Line-by-Line Breakdown	Lines 38–41 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 42–45 Method: calculateServiceCharge()

```
42      public double calculateServiceCharge(double subtotal) {
43          return Math.max(0, subtotal) * serviceChargeRate;
44      }
45
```

Component	Method: calculateServiceCharge()
Line-by-Line Breakdown	Lines 42–45 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 46–53 Method: calculateGrandTotal()

```
46      public double calculateGrandTotal(double subtotal, double discountAmount) {
47          double discount = Math.min(Math.max(0, discountAmount), subtotal);
48          double taxable = subtotal - discount;
49          double tax = calculateTax(taxable);
50          double service = calculateServiceCharge(subtotal);
51          return taxable + tax + service;
52      }
53
```

Component	Method: calculateGrandTotal()
Line-by-Line Breakdown	Lines 46–53 (8 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 54–103 **Method: generateBillReceipt()**

```
54       public String generateBillReceipt(Order order) {
55           if (order == null) return "Order not found.";
56           AppConfig config = AppConfig.getInstance();
57           StringBuilder sb = new StringBuilder();
58           sb.append("================================================\n");
59           sb.append(String.format("          %s\n", config.getProperty("restaurant.name", "Grand Feast F
ine Dining")));
60           sb.append(String.format("      %s\n", config.getProperty("restaurant.address", "104 Gourmet Bl
vd")));
61           sb.append(String.format("          Tel: %s\n", config.getProperty("restaurant.phone", "+91 987
65 43210")));
62           sb.append("================================================\n");
63           sb.append(String.format("Bill / Order No : %s\n", order.getOrderNumber()));
64           sb.append(String.format("Table           : %s\n", order.getTableNumber() != null ? order.getTa
bleNumber() : "Takeaway / Direct"));
65           sb.append(String.format("Customer        : %s\n", order.getCustomerName() != null ? order.getC
ustomerName() : "Walk-in Guest"));
66           sb.append(String.format("Date / Time     : %s\n", DateTimeUtil.formatDateTime(order.getOrderDa
teTime())));
67           sb.append("------------------------------------------------\n");
68           sb.append(String.format("%-25s %5s %9s %8s\n", "Item", "Qty", "Price", "Total"));
69           sb.append("------------------------------------------------\n");
70
71           if (order.getItems() != null) {
72               for (OrderItem item : order.getItems()) {
73                   String name = item.getItemName();
74                   if (name.length() > 24) name = name.substring(0, 21) + "...";
75                   sb.append(String.format("%-25s %5d %9.2f %8.2f\n", name, item.getQuantity(), item.getU
nitPrice(), item.getSubtotal()));
76               }
77           }
78
79           sb.append("------------------------------------------------\n");
80           sb.append(String.format("%-32s : %s\n", "Subtotal", CurrencyUtil.format(order.getSubtotal
())));
81           if (order.getDiscountAmount() > 0) {
82               sb.append(String.format("%-32s : -%s\n", "Discount Applied", CurrencyUtil.format(order.get
DiscountAmount())));
83           }
84           sb.append(String.format("%-32s : %s\n", String.format("GST (%.1f%%)", taxRate * 100), Currency
Util.format(order.getTaxAmount())));
85           sb.append(String.format("%-32s : %s\n", String.format("Service Charge (%.1f%%)", serviceCharge
Rate * 100), CurrencyUtil.format(order.getServiceCharge())));
86           sb.append("================================================\n");
87           sb.append(String.format("%-32s : %s\n", "GRAND TOTAL", CurrencyUtil.format(order.getTotalAmoun
t())));
88           sb.append("================================================\n");
89           sb.append("             THANK YOU! VISIT US AGAIN            \n");
90           sb.append("================================================\n");
91           return sb.toString();
92       }
93
94       /**
95        * FILE HANDLING — saves the generated receipt to a .txt file.
96        *
97        * Demonstrates IOException handling with try-catch:
98        * the caller receives either the saved file path (success)
99        * or a fallback message (failure).
100       *
101       * @param order the completed order
```

```
102         * @return absolute path of the saved receipt, or an error message string
103         */
```

Component	Method: generateBillReceipt()
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 54–103 (50 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 104–119` Method: saveReceiptToFile()

```
104       public String saveReceiptToFile(Order order) {
105           String receiptContent = generateBillReceipt(order);
106           BillReceiptWriter writer = new BillReceiptWriter();
107           try {
108               String savedPath = writer.saveReceipt(order, receiptContent);
109               // FILE HANDLING — log the successful save
110               ActivityLogger.getInstance().logReceiptSaved(order.getOrderNumber(), savedPath);
111               return savedPath;
112           } catch (IOException e) {
113               // EXCEPTION HANDLING — catch I/O failure; log and return a user-friendly message
114               ActivityLogger.getInstance().logError(
115                       "BillingService.saveReceiptToFile", e.getMessage());
116               return "Receipt could not be saved: " + e.getMessage();
117           }
118       }
119   }
```

Component	Method: saveReceiptToFile()
File Handling Logic	Interacts with the disk filesystem: ensures parent directory existence, opens output streams, and writes formatted text or CSV records.
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 104–119 (16 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

CustomerService.java

Path: src\main\java\com\restaurant\service\CustomerService.java | Total Lines: 82 | Logical Blocks: 9

📘 COMPONENT OVERVIEW & RESPONSIBILITY

Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–10 Package & Library Imports

```
1  package com.restaurant.service;
2
3  import com.restaurant.data.DataStructureManager;
4  import com.restaurant.model.Customer;
5  import com.restaurant.repository.CustomerRepository;
6  import com.restaurant.util.ValidationUtil;
7
8  import java.util.List;
9  import java.util.logging.Logger;
10
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Line Analysis	Covers lines 1 to 10. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 11–11 Class Declaration & Heritage

```
11  public class CustomerService {
```

Architecture Role	Defines CustomerService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 11, allowing cross-package invocation.

Lines 12–22 Class Members & Implementation

```
12      private static final Logger LOGGER = Logger.getLogger(CustomerService.class.getName());
13      private final CustomerRepository customerRepository;
14
15      public CustomerService() {
16          this.customerRepository = new CustomerRepository();
17      }
18
19      public CustomerService(CustomerRepository customerRepository) {
20          this.customerRepository = customerRepository;
21      }
22
```

Component	Class Members & Implementation
Line-by-Line Breakdown	Lines 12–22 (11 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 23–30 Method: getAllCustomers()

```
23      public List<Customer> getAllCustomers() {
24          List<Customer> list = customerRepository.findAll();
25          for (Customer c : list) {
26              DataStructureManager.getInstance().registerCustomerInIndex(c);
27          }
28          return list;
29      }
30
```

Component	Method: getAllCustomers()
Line-by-Line Breakdown	Lines 23–30 (8 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 31–34 Method: getCustomerById()

```
31      public Customer getCustomerById(int customerId) {
32          return customerRepository.findById(customerId);
33      }
34
```

Component	Method: getCustomerById()
Line-by-Line Breakdown	Lines 31–34 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 35–45 Method: getCustomerByPhone()

```
35      public Customer getCustomerByPhone(String phone) {
36          if (phone == null || phone.trim().isEmpty()) return null;
37          Customer cached = DataStructureManager.getInstance().findCustomerByPhone(phone);
38          if (cached != null) return cached;
39          Customer fromDb = customerRepository.findByPhone(phone.trim());
40          if (fromDb != null) {
41              DataStructureManager.getInstance().registerCustomerInIndex(fromDb);
42          }
43          return fromDb;
44      }
45
```

Component	Method: getCustomerByPhone()
Line-by-Line Breakdown	Lines 35–45 (11 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 46–70 Method: registerCustomer()

```
46      public boolean registerCustomer(String name, String phone, String email, String address) {
47          if (!ValidationUtil.isNotEmpty(name)) {
48              throw new IllegalArgumentException("Customer name is required.");
49          }
50          if (!ValidationUtil.isValidPhone(phone)) {
51              throw new IllegalArgumentException("Valid phone number is required.");
52          }
53          if (ValidationUtil.isNotEmpty(email) && !ValidationUtil.isValidEmail(email)) {
54              throw new IllegalArgumentException("Invalid email format.");
55          }
56
57          Customer existing = customerRepository.findByPhone(phone.trim());
58          if (existing != null) {
59              throw new IllegalArgumentException("A customer with phone " + phone + " already exists.");
60          }
61
62          Customer customer = new Customer(0, name.trim(), phone.trim(), email != null ? email.trim() :
"", address != null ? address.trim() : "");
63          boolean created = customerRepository.create(customer);
64          if (created) {
65              DataStructureManager.getInstance().registerCustomerInIndex(customer);
66              LOGGER.info("New customer registered: " + customer.getName());
67          }
68          return created;
69      }
70
```

Component	Method: registerCustomer()
Line-by-Line Breakdown	Lines 46–70 (25 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 71–78 Method: updateCustomer()

```
71      public boolean updateCustomer(Customer customer) {
72          if (customer == null || customer.getCustomerId() <= 0) return false;
73          if (!ValidationUtil.isNotEmpty(customer.getName())) {
74              throw new IllegalArgumentException("Customer name is required.");
75          }
76          return customerRepository.update(customer);
77      }
78
```

Component	Method: updateCustomer()
Line-by-Line Breakdown	Lines 71–78 (8 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 79–82 Method: deleteCustomer()

```
79      public boolean deleteCustomer(int customerId) {
80          return customerRepository.delete(customerId);
81      }
82  }
```

Component	Method: deleteCustomer()
Line-by-Line Breakdown	Lines 79–82 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

DatabaseHealthMonitor.java

Path: src\main\java\com\restaurant\service\DatabaseHealthMonitor.java | Total Lines: 173 | Logical Blocks: 7

📘 COMPONENT OVERVIEW & RESPONSIBILITY

Multi-threaded background daemon service. Manages concurrent worker queues and automated background health monitoring.

Lines 1–26 Package & Library Imports

```
1   package com.restaurant.service;
2
3   import com.restaurant.config.DatabaseConfig;
4   import com.restaurant.util.ActivityLogger;
5
6   import java.util.concurrent.Executors;
7   import java.util.concurrent.ScheduledExecutorService;
8   import java.util.concurrent.ScheduledFuture;
9   import java.util.concurrent.TimeUnit;
10  import java.util.logging.Level;
11  import java.util.logging.Logger;
12
13  /**
14   * MULTI-THREADING — Database Health Monitor
15   *
16   * Uses a ScheduledExecutorService to ping the MySQL database every 30 seconds
17   * on a dedicated background thread.  Logs a warning if connectivity is lost
18   * and attempts to reconnect automatically.
19   *
20   * Demonstrates:
21   *   - ScheduledExecutorService (Executors.newSingleThreadScheduledExecutor)
22   *   - scheduleAtFixedRate() — periodic recurring task
23   *   - Callable/Runnable submitted to an executor
24   *   - Graceful shutdown of an ExecutorService (shutdown + awaitTermination)
25   *   - Thread naming for easier debugging
26   */
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Database Integration	Imports JDBC driver contracts for executing SQL statements and connection pooling.
Concurrency Support	Multi-threading utilities (BlockingQueue, Executors, Atomic primitives) used for asynchronous tasks.
Line Analysis	Covers lines 1 to 26. Cleanly separates dependencies without wildcard star imports where specificity is required.

`Lines 27–27` **Class Declaration & Heritage**

```
27  public class DatabaseHealthMonitor {
```

Architecture Role	Defines DatabaseHealthMonitor as a core component in the Restaurant Management System.
Domain Entity	Encapsulates business entity attributes, immutable identity, and state transformations.
Access & Modifiers	Declared with public visibility on line 27, allowing cross-package invocation.

Lines 28–78 **Class Members & Implementation**

```

    private static final Logger LOGGER =
            Logger.getLogger(DatabaseHealthMonitor.class.getName());

    /** How often to ping the database (in seconds). */
    private static final long PING_INTERVAL_SECONDS = 30L;

    /** How long to wait before the first ping after startup (in seconds). */
    private static final long INITIAL_DELAY_SECONDS  = 10L;

    // Singleton
    private static DatabaseHealthMonitor instance;

    /** Single-thread scheduled executor — runs the health-check task. */
    private final ScheduledExecutorService scheduler;

    /** Handle to the scheduled future so we can cancel it if needed. */
    private ScheduledFuture<?> scheduledFuture;

    /** Tracks consecutive failures for escalated logging. */
    private int consecutiveFailures = 0;

    // ------------------------------------------------------------------
    // Singleton
    // ------------------------------------------------------------------

    private DatabaseHealthMonitor() {
        // Name the thread so it appears clearly in thread dumps
        this.scheduler = Executors.newSingleThreadScheduledExecutor(runnable -> {
            Thread t = new Thread(runnable, "DB-Health-Monitor-Thread");
            t.setDaemon(true); // allow JVM to exit without waiting for this thread
            return t;
        });
    }

    public static synchronized DatabaseHealthMonitor getInstance() {
        if (instance == null) {
            instance = new DatabaseHealthMonitor();
        }
        return instance;
    }

    // ------------------------------------------------------------------
    // Monitor lifecycle
    // ------------------------------------------------------------------

    /**
     * Starts the periodic health-check task.
     * The first check runs after INITIAL_DELAY_SECONDS seconds.
     * Subsequent checks run every PING_INTERVAL_SECONDS seconds.
     */
```

Component	Class Members & Implementation
Multi-Threading Logic	Manages worker threads, background queues, or concurrent task execution without blocking the JavaFX UI rendering loop.

Thread Synchronization	Guarantees thread-safety: prevents race conditions by allowing only one thread to execute this critical block at any instant.
Line-by-Line Breakdown	Lines 28–78 (51 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 79–101` **Method: startMonitoring()**

```
79      public void startMonitoring() {
80          if (scheduledFuture != null && !scheduledFuture.isCancelled()) {
81              LOGGER.info("DatabaseHealthMonitor is already running.");
82              return;
83          }
84
85          scheduledFuture = scheduler.scheduleAtFixedRate(
86                  this::performHealthCheck,
87                  INITIAL_DELAY_SECONDS,
88                  PING_INTERVAL_SECONDS,
89                  TimeUnit.SECONDS
90          );
91
92          LOGGER.info(String.format(
93                  "DatabaseHealthMonitor started — pinging DB every %d seconds.",
94                  PING_INTERVAL_SECONDS));
95          ActivityLogger.getInstance().logInfo("DB health monitor started.");
96      }
97
98      /**
99       * Stops monitoring and shuts down the executor cleanly.
100      * Waits up to 5 seconds for the current ping to finish before forcing shutdown.
101      */
```

Component	Method: startMonitoring()
Line-by-Line Breakdown	Lines 79–101 (23 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 102–132 **Method: stopMonitoring()**

```
102       public void stopMonitoring() {
103           if (scheduledFuture != null) {
104               scheduledFuture.cancel(false); // do not interrupt if running
105           }
106           scheduler.shutdown();
107           try {
108               if (!scheduler.awaitTermination(5, TimeUnit.SECONDS)) {
109                   scheduler.shutdownNow(); // force if it didn't finish in time
110                   LOGGER.warning("DatabaseHealthMonitor forced shutdown after timeout.");
111               }
112           } catch (InterruptedException e) {
113               Thread.currentThread().interrupt();
114               scheduler.shutdownNow();
115           }
116           LOGGER.info("DatabaseHealthMonitor stopped.");
117           ActivityLogger.getInstance().logInfo("DB health monitor stopped.");
118       }
119
120       // ------------------------------------------------------------------------
121       // The health-check task (runs on the scheduled executor thread)
122       // ------------------------------------------------------------------------
123
124       /**
125        * Performs a single database connectivity check.
126        * This method is submitted to and executed by the ScheduledExecutorService —
127        * it runs on the "DB-Health-Monitor-Thread", NOT the JavaFX thread.
128        *
129        * Exception handling: any unchecked exception must be caught here;
130        * if it escapes a ScheduledFuture task, the scheduler silently stops
131        * scheduling future runs.
132        */
```

Component	Method: stopMonitoring()
Multi-Threading Logic	Manages worker threads, background queues, or concurrent task execution without blocking the JavaFX UI rendering loop.
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 102–132 (31 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 133–169 **Method: performHealthCheck()**

```
133       private void performHealthCheck() {
134           try {
135               boolean ok = DatabaseConfig.getInstance().testConnection();
136
137               if (ok) {
138                   if (consecutiveFailures > 0) {
139                       LOGGER.info("DatabaseHealthMonitor: DB connection RESTORED after "
140                               + consecutiveFailures + " failure(s).");
141                       ActivityLogger.getInstance().logInfo("DB connection restored.");
142                   } else {
143                       LOGGER.fine("DatabaseHealthMonitor: DB ping OK.");
144                   }
145                   consecutiveFailures = 0;
146
147               } else {
148                   consecutiveFailures++;
149                   String msg = "DatabaseHealthMonitor: DB ping FAILED (consecutive failures: "
150                           + consecutiveFailures + ")";
151                   LOGGER.warning(msg);
152                   ActivityLogger.getInstance().logError("DBHealthMonitor", msg);
153
154                   if (consecutiveFailures >= 3) {
155                       LOGGER.severe("DatabaseHealthMonitor: 3+ consecutive DB failures — "
156                               + "application may be unable to serve requests.");
157                   }
158               }
159
160           } catch (Exception e) {
161               // Catch ALL exceptions — a ScheduledTask must never throw unchecked
162               consecutiveFailures++;
163               LOGGER.log(Level.SEVERE,
164                       "DatabaseHealthMonitor: unexpected error during health check", e);
165               ActivityLogger.getInstance().logError("DBHealthMonitor", e.getMessage());
166           }
167       }
168
169       /** Returns how many consecutive ping failures have been recorded. */
```

Component	Method: performHealthCheck()
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 133–169 (37 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 170–173` **Method: getConsecutiveFailures()**

```
170       public int getConsecutiveFailures() {
171           return consecutiveFailures;
172       }
173   }
```

Component	Method: getConsecutiveFailures()
Line-by-Line Breakdown	Lines 170–173 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

InventoryService.java

Path: src\main\java\com\restaurant\service\InventoryService.java | Total Lines: 80 | Logical Blocks: 11

> 📘 **COMPONENT OVERVIEW & RESPONSIBILITY**
>
> Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–10 Package & Library Imports

```
1  package com.restaurant.service;
2
3  import com.restaurant.data.SearchSortUtil;
4  import com.restaurant.exception.InsufficientStockException;
5  import com.restaurant.model.InventoryItem;
6  import com.restaurant.repository.InventoryRepository;
7  import com.restaurant.util.ValidationUtil;
8
9  import java.util.List;
10
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Line Analysis	Covers lines 1 to 10. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 11–11 Class Declaration & Heritage

```
11 public class InventoryService {
```

Architecture Role	Defines InventoryService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 11, allowing cross-package invocation.

Lines 12–21 Class Members & Implementation

```
12      private final InventoryRepository inventoryRepository;
13
14      public InventoryService() {
15          this.inventoryRepository = new InventoryRepository();
16      }
17
18      public InventoryService(InventoryRepository inventoryRepository) {
19          this.inventoryRepository = inventoryRepository;
20      }
21
```

Component	Class Members & Implementation
Line-by-Line Breakdown	Lines 12–21 (10 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 22–25 Method: getAllInventory()

```
22      public List<InventoryItem> getAllInventory() {
23          return inventoryRepository.findAll();
24      }
25
```

Component	Method: getAllInventory()
Line-by-Line Breakdown	Lines 22–25 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 26–29 Method: getLowStockItems()

```
26      public List<InventoryItem> getLowStockItems() {
27          return inventoryRepository.findLowStockItems();
28      }
29
```

Component	Method: getLowStockItems()
Line-by-Line Breakdown	Lines 26–29 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 30–40 Method: addInventoryItem()

```
30        public boolean addInventoryItem(String name, String category, double quantity, String unit, double
minStock, double unitCost, Integer supplierId) {
31            if (!ValidationUtil.isNotEmpty(name)) {
32                throw new IllegalArgumentException("Ingredient name is required.");
33            }
34            if (!ValidationUtil.isNonNegative(quantity) || !ValidationUtil.isNonNegative(minStock)) {
35                throw new IllegalArgumentException("Quantity and minimum stock cannot be negative.");
36            }
37            InventoryItem item = new InventoryItem(0, name.trim(), category, quantity, unit, minStock, uni
tCost, supplierId);
38            return inventoryRepository.create(item);
39        }
40
```

Component	Method: addInventoryItem()
Line-by-Line Breakdown	Lines 30–40 (11 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 41–47 Method: updateStockQuantity()

```
41        public boolean updateStockQuantity(int inventoryId, double newQuantity) {
42            if (!ValidationUtil.isNonNegative(newQuantity)) {
43                throw new IllegalArgumentException("Stock quantity cannot be negative.");
44            }
45            return inventoryRepository.updateStock(inventoryId, newQuantity);
46        }
47
```

Component	Method: updateStockQuantity()
Line-by-Line Breakdown	Lines 41–47 (7 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 48–65

Method: deductStock()

```
48      public void deductStock(int inventoryId, double amountToDeduct) throws InsufficientStockException
{
49          List<InventoryItem> items = getAllInventory();
50          InventoryItem target = null;
51          for (InventoryItem item : items) {
52              if (item.getInventoryId() == inventoryId) {
53                  target = item;
54                  break;
55              }
56          }
57          if (target != null) {
58              if (target.getQuantity() < amountToDeduct) {
59                  throw new InsufficientStockException("Insufficient stock for " + target.getIngredientN
ame() +
60                          ". Available: " + target.getQuantity() + " " + target.getUnit() + ", Requeste
d: " + amountToDeduct);
61              }
62              inventoryRepository.updateStock(inventoryId, target.getQuantity() - amountToDeduct);
63          }
64      }
65
```

Component	Method: deductStock()
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 48–65 (18 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 66–70

Method: updateInventoryItem()

```
66      public boolean updateInventoryItem(InventoryItem item) {
67          if (item == null || item.getInventoryId() <= 0) return false;
68          return inventoryRepository.update(item);
69      }
70
```

Component	Method: updateInventoryItem()
Line-by-Line Breakdown	Lines 66–70 (5 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 71–74

Method: deleteInventoryItem()

```
71      public boolean deleteInventoryItem(int inventoryId) {
72          return inventoryRepository.delete(inventoryId);
73      }
74
```

Component	Method: deleteInventoryItem()
Line-by-Line Breakdown	Lines 71–74 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 75–80 Method: getInventorySortedByStockAscending()

```
75      public List<InventoryItem> getInventorySortedByStockAscending() {
76          List<InventoryItem> list = getAllInventory();
77          SearchSortUtil.quickSortInventoryByQuantity(list);
78          return list;
79      }
80  }
```

Component	Method: getInventorySortedByStockAscending()
Line-by-Line Breakdown	Lines 75–80 (6 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

KitchenOrderProcessor.java

Path: src\main\java\com\restaurant\service\KitchenOrderProcessor.java | Total Lines: 205 | Logical Blocks: 9

📘 COMPONENT OVERVIEW & RESPONSIBILITY

Multi-threaded background daemon service. Manages concurrent worker queues and automated background health monitoring.

Lines 1–31 Package & Library Imports

```
1  package com.restaurant.service;
2
3  import com.restaurant.enums.OrderStatus;
4  import com.restaurant.model.Order;
5  import com.restaurant.util.ActivityLogger;
6
7  import java.util.concurrent.BlockingQueue;
8  import java.util.concurrent.LinkedBlockingQueue;
9  import java.util.concurrent.atomic.AtomicBoolean;
10 import java.util.logging.Level;
11 import java.util.logging.Logger;
12
13 /**
14  * MULTI-THREADING — Kitchen Order Processor
15  *
16  * Runs a dedicated background daemon thread that continuously polls a
17  * BlockingQueue<Order> for new kitchen orders and "processes" each one
18  * (simulates cooking by sleeping, then marks order READY).
19  *
20  * Demonstrates:
21  *  - Runnable / Thread lifecycle (start, interrupt, shutdown)
22  *  - BlockingQueue for producer-consumer pattern (thread-safe handoff)
23  *  - AtomicBoolean as a volatile shutdown flag
24  *  - InterruptedException handling inside a worker thread
25  *  - Daemon threads (does not prevent JVM exit)
26  *
27  * Usage:
28  *   KitchenOrderProcessor processor = KitchenOrderProcessor.getInstance();
29  *   processor.start();
30  *   processor.submitOrder(order);   // called by OrderService after saving
31  */
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Concurrency Support	Multi-threading utilities (BlockingQueue, Executors, Atomic primitives) used for asynchronous tasks.
Line Analysis	Covers lines 1 to 31. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 32–32 Class Declaration & Heritage

```
32  public class KitchenOrderProcessor implements Runnable {
```

Architecture Role	Defines KitchenOrderProcessor as a core component in the Restaurant Management System.
Domain Entity	Encapsulates business entity attributes, immutable identity, and state transformations.
Access & Modifiers	Declared with public visibility on line 32, allowing cross-package invocation.

Lines 33–82 **Class Members & Implementation**

```
33
34      private static final Logger LOGGER =
35              Logger.getLogger(KitchenOrderProcessor.class.getName());
36
37      /** Average simulated cooking time in milliseconds (3 seconds). */
38      private static final long COOK_TIME_MS = 3_000L;
39
40      // Singleton instance
41      private static KitchenOrderProcessor instance;
42
43      /** Thread-safe queue: OrderService puts orders in; processor takes them out. */
44      private final BlockingQueue<Order> queue = new LinkedBlockingQueue<>();
45
46      /** Volatile flag — written by main thread, read by worker thread. */
47      private final AtomicBoolean running = new AtomicBoolean(false);
48
49      /** The background worker thread. */
50      private Thread workerThread;
51
52      /** Reference to OrderService for status updates. */
53      private final OrderService orderService;
54
55      // ---------------------------------------------------------------------
56      // Singleton
57      // ---------------------------------------------------------------------
58
59      private KitchenOrderProcessor() {
60          this.orderService = new OrderService();
61      }
62
63      public static synchronized KitchenOrderProcessor getInstance() {
64          if (instance == null) {
65              instance = new KitchenOrderProcessor();
66          }
67          return instance;
68      }
69
70      // ---------------------------------------------------------------------
71      // Thread lifecycle
72      // ---------------------------------------------------------------------
73
74      /**
75       * Starts the background kitchen worker thread.
76       * Calling start() when already running is a no-op.
77       */
78      public synchronized void start() {
79          if (running.get()) {
80              LOGGER.info("KitchenOrderProcessor already running.");
81              return;
82          }
```

Component	Class Members & Implementation
Multi-Threading Logic	Manages worker threads, background queues, or concurrent task execution without blocking the JavaFX UI rendering loop.
Thread Synchronization	Guarantees thread-safety: prevents race conditions by allowing only one thread to execute this critical block at any instant.

Line-by-Line Breakdown	Lines 33–82 (50 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 83–108` Class Members & Continuation

```
83          running.set(true);
84          workerThread = new Thread(this, "Kitchen-Worker-Thread");
85          workerThread.setDaemon(true); // JVM can exit even if this thread is still waiting
86          workerThread.start();
87          LOGGER.info("KitchenOrderProcessor started — kitchen worker thread is live.");
88          ActivityLogger.getInstance().logInfo("Kitchen worker thread started.");
89      }
90
91      /**
92       * Signals the worker thread to stop after finishing the current order.
93       * Interrupts the thread if it is blocked on queue.take().
94       */
95      public synchronized void stop() {
96          running.set(false);
97          if (workerThread != null) {
98              workerThread.interrupt(); // unblock queue.take()
99          }
100         LOGGER.info("KitchenOrderProcessor stop requested.");
101     }
102
103     /**
104      * Submits an order to the processing queue.
105      * Called by OrderService after the order is persisted to the DB.
106      *
107      * @param order the newly created order to be sent to the kitchen
108      */
```

Component	Class Members & Continuation
Multi-Threading Logic	Manages worker threads, background queues, or concurrent task execution without blocking the JavaFX UI rendering loop.
Thread Synchronization	Guarantees thread-safety: prevents race conditions by allowing only one thread to execute this critical block at any instant.
Line-by-Line Breakdown	Lines 83–108 (26 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 109–116 Method: submitOrder()

```
109       public void submitOrder(Order order) {
110           if (order == null) return;
111           queue.offer(order); // non-blocking; LinkedBlockingQueue is unbounded
112           LOGGER.info("Order submitted to kitchen queue: " + order.getOrderNumber()
113                       + " | Queue size: " + queue.size());
114       }
115
116       /** Returns the number of orders currently waiting in the kitchen queue. */
```

Component	Method: submitOrder()
Multi-Threading Logic	Manages worker threads, background queues, or concurrent task execution without blocking the JavaFX UI rendering loop.
Line-by-Line Breakdown	Lines 109–116 (8 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 117–133 Method: getPendingCount()

```
117       public int getPendingCount() {
118           return queue.size();
119       }
120
121       // ------------------------------------------------------------------------
122       // Runnable — background processing logic (MULTI-THREADING)
123       // ------------------------------------------------------------------------
124
125       /**
126        * The worker thread body.
127        * Blocks on queue.take() until an order arrives, then simulates cooking
128        * and updates the order status to READY.
129        *
130        * InterruptedException is caught properly so the thread exits cleanly
131        * when stop() is called.
132        */
133       @Override
```

Component	Method: getPendingCount()
Multi-Threading Logic	Manages worker threads, background queues, or concurrent task execution without blocking the JavaFX UI rendering loop.
Line-by-Line Breakdown	Lines 117–133 (17 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 134–179 **Method: run()**

```
134       public void run() {
135           LOGGER.info("Kitchen worker thread started — waiting for orders...");
136
137           while (running.get()) {
138               try {
139                   // BLOCKING CALL: waits here until an order is available
140                   Order order = queue.take();
141
142                   LOGGER.info("Kitchen: picked up order " + order.getOrderNumber()
143                           + " — cooking started.");
144                   ActivityLogger.getInstance().logInfo(
145                           "Kitchen processing: " + order.getOrderNumber());
146
147                   // Simulate cooking time (demonstrates Thread.sleep in a worker)
148                   simulateCooking(order);
149
150                   // Mark the order as READY
151                   updateOrderReady(order);
152
153               } catch (InterruptedException e) {
154                   // Restore the interrupted flag — good practice
155                   Thread.currentThread().interrupt();
156                   LOGGER.info("Kitchen worker thread interrupted — shutting down gracefully.");
157                   break; // exit the loop and let the thread terminate
158               } catch (Exception e) {
159                   // Never let an unexpected exception kill the worker thread silently
160                   LOGGER.log(Level.SEVERE,
161                           "Unexpected error in kitchen worker thread", e);
162                   ActivityLogger.getInstance().logError(
163                           "KitchenOrderProcessor", e.getMessage());
164               }
165           }
166
167           LOGGER.info("Kitchen worker thread has stopped.");
168       }
169
170       // ------------------------------------------------------------------------
171       // Private helpers
172       // ------------------------------------------------------------------------
173
174       /**
175        * Simulates the time it takes to prepare an order.
176        * In a real system this would trigger a kitchen display update.
177        *
178        * @throws InterruptedException if the thread is interrupted during sleep
179        */
```

Component	Method: run()
Multi-Threading Logic	Manages worker threads, background queues, or concurrent task execution without blocking the JavaFX UI rendering loop.
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 134–179 (46 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 180–191 **Method: simulateCooking()**

```
180       private void simulateCooking(Order order) throws InterruptedException {
181           // Adjust cooking time slightly per item to feel realistic
182           int itemCount = (order.getItems() != null) ? order.getItems().size() : 1;
183           long cookTime = COOK_TIME_MS + (itemCount * 500L);
184
185           LOGGER.info(String.format("Kitchen: cooking order %s (%d items) for ~%d ms...",
186                   order.getOrderNumber(), itemCount, cookTime));
187
188           Thread.sleep(cookTime); // demonstrates Thread.sleep inside a worker thread
189       }
190
191       /** Marks the order as READY and logs the event. */
```

Component	Method: simulateCooking()
Multi-Threading Logic	Manages worker threads, background queues, or concurrent task execution without blocking the JavaFX UI rendering loop.
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 180–191 (12 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 192–205 **Method: updateOrderReady()**

```
192       private void updateOrderReady(Order order) {
193           try {
194               boolean updated = orderService.updateOrderStatus(order.getOrderId(), OrderStatus.READY);
195               if (updated) {
196                   LOGGER.info("Kitchen: Order " + order.getOrderNumber() + " is READY!");
197                   ActivityLogger.getInstance().logInfo(
198                           "Order READY: " + order.getOrderNumber());
199               }
200           } catch (Exception e) {
201               LOGGER.log(Level.WARNING,
202                       "Could not update order status to READY for: " + order.getOrderNumber(), e);
203           }
204       }
205   }
```

Component	Method: updateOrderReady()
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 192–205 (14 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

KitchenService.java

Path: src\main\java\com\restaurant\service\KitchenService.java | Total Lines: 36 | Logical Blocks: 7

> 📘 **COMPONENT OVERVIEW & RESPONSIBILITY**
>
> Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–9 Package & Library Imports

```
1  package com.restaurant.service;
2
3  import com.restaurant.data.DataStructureManager;
4  import com.restaurant.enums.OrderStatus;
5  import com.restaurant.model.Order;
6  import com.restaurant.repository.OrderRepository;
7
8  import java.util.List;
9
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Line Analysis	Covers lines 1 to 9. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 10–10 Class Declaration & Heritage

```
10  public class KitchenService {
```

Architecture Role	Defines KitchenService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 10, allowing cross-package invocation.

Lines 11–20 Class Members & Implementation

```
11      private final OrderRepository orderRepository;
12
13      public KitchenService() {
14          this.orderRepository = new OrderRepository();
15      }
16
17      public KitchenService(OrderRepository orderRepository) {
18          this.orderRepository = orderRepository;
19      }
20
```

Component	Class Members & Implementation
Line-by-Line Breakdown	Lines 11–20 (10 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 21–24 Method: getActiveKitchenOrders()

```
21      public List<Order> getActiveKitchenOrders() {
22          return orderRepository.findActiveKitchenOrders();
23      }
24
```

Component	Method: getActiveKitchenOrders()
Line-by-Line Breakdown	Lines 21–24 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 25–28 Method: markAsPreparing()

```
25      public boolean markAsPreparing(int orderId) {
26          return orderRepository.updateStatus(orderId, OrderStatus.PREPARING);
27      }
28
```

Component	Method: markAsPreparing()
Line-by-Line Breakdown	Lines 25–28 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 29–32 Method: markAsReady()

```
29      public boolean markAsReady(int orderId) {
30          return orderRepository.updateStatus(orderId, OrderStatus.READY);
31      }
32
```

Component	Method: markAsReady()
Line-by-Line Breakdown	Lines 29–32 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 33–36` **Method: markAsServed()**

```
33        public boolean markAsServed(int orderId) {
34            return orderRepository.updateStatus(orderId, OrderStatus.SERVED);
35        }
36    }
```

Component	Method: markAsServed()
Line-by-Line Breakdown	Lines 33–36 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

MenuService.java

Path: src\main\java\com\restaurant\service\MenuService.java | Total Lines: 95 | Logical Blocks: 11

> 📘 **COMPONENT OVERVIEW & RESPONSIBILITY**
>
> Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–13 Package & Library Imports

```
1   package com.restaurant.service;
2
3   import com.restaurant.data.DataStructureManager;
4   import com.restaurant.data.SearchSortUtil;
5   import com.restaurant.exception.MenuItemNotAvailableException;
6   import com.restaurant.model.MenuCategory;
7   import com.restaurant.model.MenuItem;
8   import com.restaurant.repository.MenuRepository;
9   import com.restaurant.util.ValidationUtil;
10
11  import java.util.ArrayList;
12  import java.util.List;
13
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Line Analysis	Covers lines 1 to 13. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 14–14 Class Declaration & Heritage

```
14  public class MenuService {
```

Architecture Role	Defines MenuService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 14, allowing cross-package invocation.

Lines 15–24 Class Members & Implementation

```
15      private final MenuRepository menuRepository;
16
17      public MenuService() {
18          this.menuRepository = new MenuRepository();
19      }
20
21      public MenuService(MenuRepository menuRepository) {
22          this.menuRepository = menuRepository;
23      }
24
```

Component	Class Members & Implementation
Line-by-Line Breakdown	Lines 15–24 (10 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 25–28 Method: getAllCategories()

```
25      public List<MenuCategory> getAllCategories() {
26          return menuRepository.findAllCategories();
27      }
28
```

Component	Method: getAllCategories()
Line-by-Line Breakdown	Lines 25–28 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 29–34 Method: getAllMenuItems()

```
29      public List<MenuItem> getAllMenuItems() {
30          List<MenuItem> items = menuRepository.findAllItems();
31          DataStructureManager.getInstance().cacheMenuItems(items);
32          return items;
33      }
34
```

Component	Method: getAllMenuItems()
Line-by-Line Breakdown	Lines 29–34 (6 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 35–40 Method: getMenuItemById()

```
35      public MenuItem getMenuItemById(int itemId) {
36          MenuItem cached = DataStructureManager.getInstance().getMenuItemById(itemId);
37          if (cached != null) return cached;
38          return menuRepository.findItemById(itemId);
39      }
40
```

Component	Method: getMenuItemById()
Line-by-Line Breakdown	Lines 35–40 (6 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 41–50 Method: verifyItemAvailable()

```
41      public void verifyItemAvailable(int itemId) throws MenuItemNotAvailableException {
42          MenuItem item = getMenuItemById(itemId);
43          if (item == null) {
44              throw new MenuItemNotAvailableException("Menu item #" + itemId + " was not found.");
45          }
46          if (!item.isAvailable()) {
47              throw new MenuItemNotAvailableException("Menu item '" + item.getName() + "' is currently m
arked as OUT OF STOCK.");
48          }
49      }
50
```

Component	Method: verifyItemAvailable()
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 41–50 (10 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 51–61 Method: addMenuItem()

```
51      public boolean addMenuItem(int categoryId, String name, String description, double price, int prep
Time, boolean isAvailable) {
52          if (!ValidationUtil.isNotEmpty(name)) {
53              throw new IllegalArgumentException("Menu item name is required.");
54          }
55          if (!ValidationUtil.isValidPrice(price)) {
56              throw new IllegalArgumentException("Price cannot be negative.");
57          }
58          MenuItem item = new MenuItem(0, categoryId, name.trim(), description, price, prepTime, isAvail
able);
59          return menuRepository.createItem(item);
60      }
61
```

Component	Method: addMenuItem()
Line-by-Line Breakdown	Lines 51–61 (11 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 62–69 Method: updateMenuItem()

```
62      public boolean updateMenuItem(MenuItem item) {
63          if (item == null || item.getItemId() <= 0) return false;
64          if (!ValidationUtil.isValidPrice(item.getPrice())) {
65              throw new IllegalArgumentException("Price cannot be negative.");
66          }
67          return menuRepository.updateItem(item);
68      }
69
```

Component	Method: updateMenuItem()
Line-by-Line Breakdown	Lines 62–69 (8 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 70–73 Method: deleteMenuItem()

```
70      public boolean deleteMenuItem(int itemId) {
71          return menuRepository.deleteItem(itemId);
72      }
73
```

Component	Method: deleteMenuItem()
Line-by-Line Breakdown	Lines 70–73 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 74–95

Method: searchAndFilter()

```
74        public List<MenuItem> searchAndFilter(String query, Integer categoryId, boolean sortPriceAsc) {
75            List<MenuItem> all = getAllMenuItems();
76            List<MenuItem> filtered = new ArrayList<>();
77
78            for (MenuItem item : all) {
79                boolean matchesCat = (categoryId == null || categoryId == 0 || item.getCategoryId() == cat
egoryId);
80                boolean matchesQuery = (query == null || query.trim().isEmpty() ||
81                        item.getName().toLowerCase().contains(query.toLowerCase().trim()) ||
82                        (item.getDescription() != null && item.getDescription().toLowerCase().contains(que
ry.toLowerCase().trim())));
83
84                if (matchesCat && matchesQuery) {
85                    filtered.add(item);
86                }
87            }
88
89            if (sortPriceAsc) {
90                SearchSortUtil.quickSortMenuItemsByPrice(filtered);
91            }
92
93            return filtered;
94        }
95    }
```

Component	Method: searchAndFilter()
Line-by-Line Breakdown	Lines 74–95 (22 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

OrderService.java

Path: src\main\java\com\restaurant\service\OrderService.java | Total Lines: 110 | Logical Blocks: 9

> 📘 **COMPONENT OVERVIEW & RESPONSIBILITY**
>
> Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–18 Package & Library Imports

```
1  package com.restaurant.service;
2
3  import com.restaurant.data.DataStructureManager;
4  import com.restaurant.enums.OrderStatus;
5  import com.restaurant.enums.TableStatus;
6  import com.restaurant.exception.InvalidOrderException;
7  import com.restaurant.exception.MenuItemNotAvailableException;
8  import com.restaurant.exception.TableNotAvailableException;
9  import com.restaurant.model.Order;
10 import com.restaurant.model.OrderItem;
11 import com.restaurant.repository.OrderRepository;
12 import com.restaurant.repository.TableRepository;
13 import com.restaurant.util.ActivityLogger;
14
15 import java.time.LocalDateTime;
16 import java.util.List;
17 import java.util.logging.Logger;
18
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Line Analysis	Covers lines 1 to 18. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 19–19 Class Declaration & Heritage

```
19 public class OrderService {
```

Architecture Role	Defines OrderService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 19, allowing cross-package invocation.

`Lines 20–36` **Class Members & Implementation**

```
20        private static final Logger LOGGER = Logger.getLogger(OrderService.class.getName());
21        private final OrderRepository orderRepository;
22        private final TableRepository tableRepository;
23        private final MenuService menuService;
24
25        public OrderService() {
26            this.orderRepository = new OrderRepository();
27            this.tableRepository = new TableRepository();
28            this.menuService = new MenuService();
29        }
30
31        public OrderService(OrderRepository orderRepository, TableRepository tableRepository, MenuService
menuService) {
32            this.orderRepository = orderRepository;
33            this.tableRepository = tableRepository;
34            this.menuService = menuService;
35        }
36
```

Component	Class Members & Implementation
Line-by-Line Breakdown	Lines 20–36 (17 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 37–89 **Method: createNewOrder()**

```
37       public Order createNewOrder(Integer customerId, Integer tableId, int employeeId, List<OrderItem> i
tems, String specialNotes)
38               throws InvalidOrderException, TableNotAvailableException, MenuItemNotAvailableException {
39
40           if (items == null || items.isEmpty()) {
41               throw new InvalidOrderException("Order must contain at least one menu item.");
42           }
43
44           if (tableId != null && tableId > 0) {
45               var table = tableRepository.findById(tableId);
46               if (table != null && table.getStatus() == TableStatus.OCCUPIED) {
47                   // Table occupied check
48               }
49           }
50
51           // Validate all menu items
52           for (OrderItem item : items) {
53               if (item.getQuantity() <= 0) {
54                   throw new InvalidOrderException("Quantity for item '" + item.getItemName() + "' must b
e positive.");
55               }
56               menuService.verifyItemAvailable(item.getItemId());
57           }
58
59           String orderNumber = "ORD-" + System.currentTimeMillis() % 10000000;
60           Order order = new Order(orderNumber, customerId, tableId, employeeId);
61           order.setSpecialNotes(specialNotes);
62           order.setItems(items);
63           order.setStatus(OrderStatus.CONFIRMED);
64
65           boolean saved = orderRepository.createOrderWithItems(order);
66           if (!saved) {
67               throw new InvalidOrderException("Database operation failed while saving the order.");
68           }
69
70           // Update table status to OCCUPIED
71           if (tableId != null && tableId > 0) {
72               tableRepository.updateStatus(tableId, TableStatus.OCCUPIED);
73           }
74
75           // Enqueue into kitchen priority queue (legacy DSA structure)
76           DataStructureManager.getInstance().getKitchenOrderQueue().enqueue(order, 5);
77
78           // MULTI-THREADING — submit order to background kitchen processor thread
79           com.restaurant.service.KitchenOrderProcessor.getInstance().submitOrder(order);
80
81           // FILE HANDLING — log the new order creation event
82           ActivityLogger.getInstance().logOrderCreated(
83                   order.getOrderNumber(),
84                   order.getTableNumber() != null ? order.getTableNumber() : "Takeaway",
85                   order.getTotalAmount());
86
87           LOGGER.info("Order created successfully: " + order.getOrderNumber() + " with total: " + order.
getTotalAmount());
88           return order;
89       }
```

Component	Method: createNewOrder()

Multi-Threading Logic	Manages worker threads, background queues, or concurrent task execution without blocking the JavaFX UI rendering loop.
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 37–89 (53 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 90–90` Class Members & Continuation

```
90
```

Component	Class Members & Continuation
Line-by-Line Breakdown	Lines 90–90 (1 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 91–94` Method: getAllOrders()

```
91      public List<Order> getAllOrders() {
92          return orderRepository.findAll();
93      }
94
```

Component	Method: getAllOrders()
Line-by-Line Breakdown	Lines 91–94 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 95–98` Method: getOrderById()

```
95      public Order getOrderById(int orderId) {
96          return orderRepository.findById(orderId);
97      }
98
```

Component	Method: getOrderById()
Line-by-Line Breakdown	Lines 95–98 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 99–102` Method: updateOrderStatus()

```
99      public boolean updateOrderStatus(int orderId, OrderStatus status) {
100         return orderRepository.updateStatus(orderId, status);
101     }
102
```

Component	Method: updateOrderStatus()
Line-by-Line Breakdown	Lines 99–102 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 103–110` **Method: cancelOrder()**

```
103       public boolean cancelOrder(int orderId) {
104           Order order = orderRepository.findById(orderId);
105           if (order != null && order.getTableId() != null) {
106               tableRepository.updateStatus(order.getTableId(), TableStatus.AVAILABLE);
107           }
108           return orderRepository.updateStatus(orderId, OrderStatus.CANCELLED);
109       }
110   }
```

Component	Method: cancelOrder()
Line-by-Line Breakdown	Lines 103–110 (8 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

PaymentService.java

Path: src\main\java\com\restaurant\service\PaymentService.java | Total Lines: 86 | Logical Blocks: 6

📘 COMPONENT OVERVIEW & RESPONSIBILITY

Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–18 Package & Library Imports

```
1   package com.restaurant.service;
2
3   import com.restaurant.enums.OrderStatus;
4   import com.restaurant.enums.PaymentMethod;
5   import com.restaurant.enums.PaymentStatus;
6   import com.restaurant.enums.TableStatus;
7   import com.restaurant.exception.PaymentFailedException;
8   import com.restaurant.model.Customer;
9   import com.restaurant.model.Order;
10  import com.restaurant.model.Payment;
11  import com.restaurant.repository.CustomerRepository;
12  import com.restaurant.repository.OrderRepository;
13  import com.restaurant.repository.PaymentRepository;
14  import com.restaurant.repository.TableRepository;
15
16  import java.util.List;
17  import java.util.logging.Logger;
18
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Line Analysis	Covers lines 1 to 18. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 19–19 Class Declaration & Heritage

```
19  public class PaymentService {
```

Architecture Role	Defines PaymentService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 19, allowing cross-package invocation.

Lines 20–43 **Class Members & Implementation**

```
20      private static final Logger LOGGER = Logger.getLogger(PaymentService.class.getName());
21      private final PaymentRepository paymentRepository;
22      private final OrderRepository orderRepository;
23      private final TableRepository tableRepository;
24      private final CustomerRepository customerRepository;
25
26      public PaymentService() {
27          this.paymentRepository = new PaymentRepository();
28          this.orderRepository = new OrderRepository();
29          this.tableRepository = new TableRepository();
30          this.customerRepository = new CustomerRepository();
31      }
32
33      public PaymentService(PaymentRepository paymentRepository, OrderRepository orderRepository,
34                            TableRepository tableRepository, CustomerRepository customerRepository) {
35          this.paymentRepository = paymentRepository;
36          this.orderRepository = orderRepository;
37          this.tableRepository = tableRepository;
38          this.customerRepository = customerRepository;
39      }
40
41      /**
42       * Executes the payment polymorphically and finalizes the order workflow.
43       */
```

Component	Class Members & Implementation
Line-by-Line Breakdown	Lines 20–43 (24 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 44–78 **Method: processAndFinalizePayment()**

```
44        public boolean processAndFinalizePayment(Payment payment, Order order) throws PaymentFailedExcepti
on {
45            if (payment == null || order == null) {
46                throw new PaymentFailedException("Invalid payment or order data.");
47            }
48
49            // Execute Polymorphic payment processing
50            boolean success = payment.processPayment();
51            if (!success || payment.getStatus() != PaymentStatus.PAID) {
52                throw new PaymentFailedException("Payment processing could not be completed.");
53            }
54
55            // Persist payment transaction
56            paymentRepository.recordPayment(payment);
57
58            // Mark order as COMPLETED
59            orderRepository.updateStatus(order.getOrderId(), OrderStatus.COMPLETED);
60
61            // Release table -> CLEANING
62            if (order.getTableId() != null && order.getTableId() > 0) {
63                tableRepository.updateStatus(order.getTableId(), TableStatus.CLEANING);
64            }
65
66            // Update customer spending and loyalty points
67            if (order.getCustomerId() != null && order.getCustomerId() > 0) {
68                Customer customer = customerRepository.findById(order.getCustomerId());
69                if (customer != null) {
70                    customer.addSpending(payment.getAmount());
71                    customerRepository.update(customer);
72                }
73            }
74
75            LOGGER.info("Payment finalized successfully for Order: " + order.getOrderNumber() + ", Inv: "
+ payment.getInvoiceNumber());
76            return true;
77        }
78
```

Component	Method: processAndFinalizePayment()
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 44–78 (35 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 79–82 Method: getPaymentForOrder()

```
79        public Payment getPaymentForOrder(int orderId) {
80            return paymentRepository.findByOrderId(orderId);
81        }
82
```

Component	Method: getPaymentForOrder()
Line-by-Line Breakdown	Lines 79–82 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 83–86 Method: getAllPayments()

```
83        public List<Payment> getAllPayments() {
84            return paymentRepository.findAll();
85        }
86    }
```

Component	Method: getAllPayments()
Line-by-Line Breakdown	Lines 83–86 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

ReportService.java

Path: src\main\java\com\restaurant\service\ReportService.java | Total Lines: 127 | Logical Blocks: 5

> 📘 **COMPONENT OVERVIEW & RESPONSIBILITY**
>
> Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–13 Package & Library Imports

```
1   package com.restaurant.service;
2
3   import com.restaurant.config.DatabaseConfig;
4   import com.restaurant.exception.DatabaseOperationException;
5
6   import java.sql.Connection;
7   import java.sql.PreparedStatement;
8   import java.sql.ResultSet;
9   import java.sql.SQLException;
10  import java.util.HashMap;
11  import java.util.LinkedHashMap;
12  import java.util.Map;
13
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Database Integration	Imports JDBC driver contracts for executing SQL statements and connection pooling.
Line Analysis	Covers lines 1 to 13. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 14–14 Class Declaration & Heritage

```
14  public class ReportService {
```

Architecture Role	Defines ReportService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 14, allowing cross-package invocation.

Lines 15–60 **Class Members & Implementation**

```
15      private final DatabaseConfig dbConfig = DatabaseConfig.getInstance();
16
17      public Map<String, Double> getSalesByCategory() {
18          Map<String, Double> map = new LinkedHashMap<>();
19          String sql = "SELECT c.name as category_name, SUM(oi.subtotal) as total_sales " +
20                       "FROM order_items oi " +
21                       "JOIN menu_items m ON oi.item_id = m.item_id " +
22                       "JOIN menu_categories c ON m.category_id = c.category_id " +
23                       "JOIN orders o ON oi.order_id = o.order_id " +
24                       "WHERE o.status = 'COMPLETED' " +
25                       "GROUP BY c.category_id, c.name " +
26                       "ORDER BY total_sales DESC";
27          try (Connection conn = dbConfig.getConnection();
28               PreparedStatement stmt = conn.prepareStatement(sql);
29               ResultSet rs = stmt.executeQuery()) {
30              while (rs.next()) {
31                  map.put(rs.getString("category_name"), rs.getDouble("total_sales"));
32              }
33          } catch (SQLException e) {
34              throw new DatabaseOperationException("Error generating sales by category report", e);
35          }
36          return map;
37      }
38
39      public Map<String, Integer> getMostPopularItems(int limit) {
40          Map<String, Integer> map = new LinkedHashMap<>();
41          String sql = "SELECT m.name as item_name, SUM(oi.quantity) as total_quantity " +
42                       "FROM order_items oi " +
43                       "JOIN menu_items m ON oi.item_id = m.item_id " +
44                       "JOIN orders o ON oi.order_id = o.order_id " +
45                       "WHERE o.status = 'COMPLETED' " +
46                       "GROUP BY m.item_id, m.name " +
47                       "ORDER BY total_quantity DESC LIMIT ?";
48          try (Connection conn = dbConfig.getConnection();
49               PreparedStatement stmt = conn.prepareStatement(sql)) {
50              stmt.setInt(1, limit);
51              try (ResultSet rs = stmt.executeQuery()) {
52                  while (rs.next()) {
53                      map.put(rs.getString("item_name"), rs.getInt("total_quantity"));
54                  }
55              }
56          } catch (SQLException e) {
57              throw new DatabaseOperationException("Error generating popular items report", e);
58          }
59          return map;
60      }
```

Component	Class Members & Implementation
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 15–60 (46 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 61–108` **Class Members & Continuation**

```
61
62      public Map<String, Double> getPaymentMethodBreakdown() {
63          Map<String, Double> map = new LinkedHashMap<>();
64          String sql = "SELECT payment_method, SUM(amount) as total_amount FROM payments " +
65                       "WHERE status = 'PAID' GROUP BY payment_method";
66          try (Connection conn = dbConfig.getConnection();
67               PreparedStatement stmt = conn.prepareStatement(sql);
68               ResultSet rs = stmt.executeQuery()) {
69              while (rs.next()) {
70                  map.put(rs.getString("payment_method"), rs.getDouble("total_amount"));
71              }
72          } catch (SQLException e) {
73              throw new DatabaseOperationException("Error generating payment methods report", e);
74          }
75          return map;
76      }
77
78      public Map<String, Object> getTodaySummaryMetrics() {
79          Map<String, Object> metrics = new HashMap<>();
80          String sqlRevenue = "SELECT COALESCE(SUM(total_amount), 0) as today_rev, COUNT(*) as today_ord
ers " +
81                              "FROM orders WHERE DATE(order_date_time) = CURDATE() AND status = 'COMPLETE
D'";
82          String sqlTables = "SELECT " +
83                             "SUM(CASE WHEN status = 'AVAILABLE' THEN 1 ELSE 0 END) as avail_tables, " +
84                             "SUM(CASE WHEN status = 'OCCUPIED' THEN 1 ELSE 0 END) as occ_tables " +
85                             "FROM restaurant_tables";
86          String sqlKitchen = "SELECT COUNT(*) as pending_kitchen FROM orders WHERE status IN ('CONFIRME
D', 'PREPARING')";
87          String sqlLowStock = "SELECT COUNT(*) as low_stock_count FROM inventory WHERE quantity <= mini
mum_stock";
88
89          try (Connection conn = dbConfig.getConnection()) {
90              try (PreparedStatement s1 = conn.prepareStatement(sqlRevenue);
91                   ResultSet rs1 = s1.executeQuery()) {
92                  if (rs1.next()) {
93                      metrics.put("todayRevenue", rs1.getDouble("today_rev"));
94                      metrics.put("todayOrders", rs1.getInt("today_orders"));
95                  }
96              }
97              try (PreparedStatement s2 = conn.prepareStatement(sqlTables);
98                   ResultSet rs2 = s2.executeQuery()) {
99                  if (rs2.next()) {
100                     metrics.put("availableTables", rs2.getInt("avail_tables"));
101                     metrics.put("occupiedTables", rs2.getInt("occ_tables"));
102                 }
103             }
104             try (PreparedStatement s3 = conn.prepareStatement(sqlKitchen);
105                  ResultSet rs3 = s3.executeQuery()) {
106                 if (rs3.next()) {
107                     metrics.put("pendingKitchen", rs3.getInt("pending_kitchen"));
108                 }
```

Component	Class Members & Continuation
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.

Line-by-Line Breakdown	Lines 61–108 (48 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 109–127` Class Members & Continuation

```
109             }
110             try (PreparedStatement s4 = conn.prepareStatement(sqlLowStock);
111                  ResultSet rs4 = s4.executeQuery()) {
112                 if (rs4.next()) {
113                     metrics.put("lowStockCount", rs4.getInt("low_stock_count"));
114                 }
115             }
116         } catch (SQLException e) {
117             // Provide sensible fallback if query cannot run yet
118             metrics.put("todayRevenue", 2602.75);
119             metrics.put("todayOrders", 2);
120             metrics.put("availableTables", 8);
121             metrics.put("occupiedTables", 2);
122             metrics.put("pendingKitchen", 0);
123             metrics.put("lowStockCount", 1);
124         }
125         return metrics;
126     }
127 }
```

Component	Class Members & Continuation
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 109–127 (19 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

ReservationService.java

Path: src\main\java\com\restaurant\service\ReservationService.java | Total Lines: 70 | Logical Blocks: 6

> 📘 **COMPONENT OVERVIEW & RESPONSIBILITY**
>
> Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–14 Package & Library Imports

```
1  package com.restaurant.service;
2
3  import com.restaurant.enums.ReservationStatus;
4  import com.restaurant.enums.TableStatus;
5  import com.restaurant.exception.InvalidReservationException;
6  import com.restaurant.model.Reservation;
7  import com.restaurant.model.RestaurantTable;
8  import com.restaurant.repository.ReservationRepository;
9  import com.restaurant.repository.TableRepository;
10
11 import java.time.LocalDate;
12 import java.time.LocalTime;
13 import java.util.List;
14
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Line Analysis	Covers lines 1 to 14. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 15–15 Class Declaration & Heritage

```
15 public class ReservationService {
```

Architecture Role	Defines ReservationService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 15, allowing cross-package invocation.

Lines 16–28 Class Members & Implementation

```
16        private final ReservationRepository reservationRepository;
17        private final TableRepository tableRepository;
18
19        public ReservationService() {
20            this.reservationRepository = new ReservationRepository();
21            this.tableRepository = new TableRepository();
22        }
23
24        public ReservationService(ReservationRepository reservationRepository, TableRepository tableReposi
tory) {
25            this.reservationRepository = reservationRepository;
26            this.tableRepository = tableRepository;
27        }
28
```

Component	Class Members & Implementation
Line-by-Line Breakdown	Lines 16–28 (13 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 29–32 Method: getAllReservations()

```
29        public List<Reservation> getAllReservations() {
30            return reservationRepository.findAll();
31        }
32
```

Component	Method: getAllReservations()
Line-by-Line Breakdown	Lines 29–32 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 33–66

Method: createReservation()

```
33       public boolean createReservation(int customerId, int tableId, LocalDate date, LocalTime time, int
guestCount, String specialRequest)
34               throws InvalidReservationException {
35
36           if (date == null || date.isBefore(LocalDate.now())) {
37               throw new InvalidReservationException("Reservation date cannot be in the past.");
38           }
39           if (guestCount <= 0) {
40               throw new InvalidReservationException("Guest count must be at least 1.");
41           }
42
43           RestaurantTable table = tableRepository.findById(tableId);
44           if (table == null) {
45               throw new InvalidReservationException("Selected table not found.");
46           }
47           if (guestCount > table.getCapacity() + 1) {
48               throw new InvalidReservationException("Guest count exceeds table capacity (" + table.getCa
pacity() + ").");
49           }
50
51           // Conflict check: prevent double booking within a 2-hour window
52           if (reservationRepository.hasConflict(tableId, date, time, 0)) {
53               throw new InvalidReservationException("Table " + table.getTableNumber() + " already has a
reservation booked at that time.");
54           }
55
56           Reservation res = new Reservation(customerId, tableId, date, time, guestCount);
57           res.setSpecialRequest(specialRequest);
58           res.setStatus(ReservationStatus.CONFIRMED);
59
60           boolean created = reservationRepository.create(res);
61           if (created && date.equals(LocalDate.now())) {
62               tableRepository.updateStatus(tableId, TableStatus.RESERVED);
63           }
64           return created;
65       }
66
```

Component	Method: createReservation()
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 33–66 (34 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

`Lines 67–70` Method: updateStatus()

```
67      public boolean updateStatus(int reservationId, ReservationStatus status) {
68          return reservationRepository.updateStatus(reservationId, status);
69      }
70  }
```

Component	Method: updateStatus()
Line-by-Line Breakdown	Lines 67–70 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

SupplierService.java

Path: src\main\java\com\restaurant\service\SupplierService.java | Total Lines: 47 | Logical Blocks: 8

> 📘 **COMPONENT OVERVIEW & RESPONSIBILITY**
>
> Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–8 Package & Library Imports

```
1  package com.restaurant.service;
2
3  import com.restaurant.model.Supplier;
4  import com.restaurant.repository.SupplierRepository;
5  import com.restaurant.util.ValidationUtil;
6
7  import java.util.List;
8
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Line Analysis	Covers lines 1 to 8. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 9–9 Class Declaration & Heritage

```
9  public class SupplierService {
```

Architecture Role	Defines SupplierService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 9, allowing cross-package invocation.

Lines 10–19 Class Members & Implementation

```
10      private final SupplierRepository supplierRepository;
11
12      public SupplierService() {
13          this.supplierRepository = new SupplierRepository();
14      }
15
16      public SupplierService(SupplierRepository supplierRepository) {
17          this.supplierRepository = supplierRepository;
18      }
19
```

Component	Class Members & Implementation
Line-by-Line Breakdown	Lines 10–19 (10 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 20–23 Method: getAllSuppliers()

```
20      public List<Supplier> getAllSuppliers() {
21          return supplierRepository.findAll();
22      }
23
```

Component	Method: getAllSuppliers()
Line-by-Line Breakdown	Lines 20–23 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 24–27 Method: getSupplierById()

```
24      public Supplier getSupplierById(int supplierId) {
25          return supplierRepository.findById(supplierId);
26      }
27
```

Component	Method: getSupplierById()
Line-by-Line Breakdown	Lines 24–27 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 28–38 Method: addSupplier()

```
28        public boolean addSupplier(String name, String contactPerson, String phone, String email, String a
ddress, String suppliedCategory) {
29            if (!ValidationUtil.isNotEmpty(name)) {
30                throw new IllegalArgumentException("Supplier company name is required.");
31            }
32            if (!ValidationUtil.isValidPhone(phone)) {
33                throw new IllegalArgumentException("Valid contact phone number is required.");
34            }
35            Supplier s = new Supplier(0, name.trim(), contactPerson, phone.trim(), email, address, supplie
dCategory);
36            return supplierRepository.create(s);
37        }
38
```

Component	Method: addSupplier()
Line-by-Line Breakdown	Lines 28–38 (11 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 39–43 Method: updateSupplier()

```
39        public boolean updateSupplier(Supplier s) {
40            if (s == null || s.getSupplierId() <= 0) return false;
41            return supplierRepository.update(s);
42        }
43
```

Component	Method: updateSupplier()
Line-by-Line Breakdown	Lines 39–43 (5 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 44–47 Method: deleteSupplier()

```
44        public boolean deleteSupplier(int supplierId) {
45            return supplierRepository.delete(supplierId);
46        }
47    }
```

Component	Method: deleteSupplier()
Line-by-Line Breakdown	Lines 44–47 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

TableService.java

Path: src\main\java\com\restaurant\service\TableService.java | Total Lines: 63 | Logical Blocks: 10

> 📘 **COMPONENT OVERVIEW & RESPONSIBILITY**
>
> Business Service Layer component. Orchestrates domain transactions, enforces business constraints, and coordinates data persistence.

Lines 1–10 Package & Library Imports

```
1   package com.restaurant.service;
2
3   import com.restaurant.enums.TableStatus;
4   import com.restaurant.exception.TableNotAvailableException;
5   import com.restaurant.model.RestaurantTable;
6   import com.restaurant.repository.TableRepository;
7   import com.restaurant.util.ValidationUtil;
8
9   import java.util.List;
10
```

Purpose	Defines the namespace package and imports essential Java standard libraries (collections, concurrency, I/O, SQL) and JavaFX UI components.
Line Analysis	Covers lines 1 to 10. Cleanly separates dependencies without wildcard star imports where specificity is required.

Lines 11–11 Class Declaration & Heritage

```
11  public class TableService {
```

Architecture Role	Defines TableService as a core component in the Restaurant Management System.
Service Layer	Encapsulates business workflows, transaction boundaries, validation rules, and inter-thread dispatches.
Access & Modifiers	Declared with public visibility on line 11, allowing cross-package invocation.

Lines 12–21 Class Members & Implementation

```
12      private final TableRepository tableRepository;
13
14      public TableService() {
15          this.tableRepository = new TableRepository();
16      }
17
18      public TableService(TableRepository tableRepository) {
19          this.tableRepository = tableRepository;
20      }
21
```

Component	Class Members & Implementation
Line-by-Line Breakdown	Lines 12–21 (10 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 22–25 Method: getAllTables()

```
22      public List<RestaurantTable> getAllTables() {
23          return tableRepository.findAll();
24      }
25
```

Component	Method: getAllTables()
Line-by-Line Breakdown	Lines 22–25 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 26–29 Method: getTableById()

```
26      public RestaurantTable getTableById(int tableId) {
27          return tableRepository.findById(tableId);
28      }
29
```

Component	Method: getTableById()
Line-by-Line Breakdown	Lines 26–29 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 30–40 Method: addTable()

```
30      public boolean addTable(String tableNumber, int capacity, String location) {
31          if (!ValidationUtil.isNotEmpty(tableNumber)) {
32              throw new IllegalArgumentException("Table number cannot be empty.");
33          }
34          if (!ValidationUtil.isValidCapacity(capacity)) {
35              throw new IllegalArgumentException("Table capacity must be between 1 and 100.");
36          }
37          RestaurantTable table = new RestaurantTable(0, tableNumber.trim(), capacity, location, TableSt
atus.AVAILABLE);
38          return tableRepository.create(table);
39      }
40
```

Component	Method: addTable()
Line-by-Line Breakdown	Lines 30–40 (11 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 41–45 Method: updateTable()

```
41      public boolean updateTable(RestaurantTable table) {
42          if (table == null || table.getTableId() <= 0) return false;
43          return tableRepository.update(table);
44      }
45
```

Component	Method: updateTable()
Line-by-Line Breakdown	Lines 41–45 (5 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 46–49 Method: changeTableStatus()

```
46      public boolean changeTableStatus(int tableId, TableStatus status) {
47          return tableRepository.updateStatus(tableId, status);
48      }
49
```

Component	Method: changeTableStatus()
Line-by-Line Breakdown	Lines 46–49 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 50–59 Method: verifyTableAvailability()

```
50        public void verifyTableAvailability(int tableId) throws TableNotAvailableException {
51            RestaurantTable table = tableRepository.findById(tableId);
52            if (table == null) {
53                throw new TableNotAvailableException("Table #" + tableId + " does not exist.");
54            }
55            if (table.getStatus() == TableStatus.OCCUPIED) {
56                throw new TableNotAvailableException("Table " + table.getTableNumber() + " is currently OC
CUPIED.");
57            }
58        }
59
```

Component	Method: verifyTableAvailability()
Exception Handling	Implements robust error handling (try-with-resources or try-catch). Automatically closes database or file stream handles, eliminating resource leaks.
Line-by-Line Breakdown	Lines 50–59 (10 lines): Implements operational logic, field assignments, validation checks, and event dispatches.

Lines 60–63 Method: deleteTable()

```
60        public boolean deleteTable(int tableId) {
61            return tableRepository.delete(tableId);
62        }
63    }
```

Component	Method: deleteTable()
Line-by-Line Breakdown	Lines 60–63 (4 lines): Implements operational logic, field assignments, validation checks, and event dispatches.